Exp:6 | Error detection and correction using Hamming code

Date: 5/5/26

Aim:

To write a program to implement error detection and correction using the hamming code

Algorithm:

1) Start the program
2) Read the number of 4 digits data blocks
3) For each digits 4bits data block(s) perform hamming

$P_1 = d_1 \text{ XOR } d_2 \text{ XOR } d_4$

$P_2 = d_1 \text{ XOR } d_3 \text{ XOR } d_4$

$P_3 = d_2 \text{ XOR } d_3 \text{ XOR } d_4$

4) Simulate transmission by introducing a single bit error
5) Compute error position = $c_4c_2c_1$
6) If error position = no error has occured
7) Extract the orgin data bits from the corrected bit.
8) Stop the program

program:

```
def encoded(A):
    h = [0]*7
    h[2] = d[0]
    h[4] = d[1]
    h[5] = d[2]
    h[6] = d[3]
    h[0] = h[2] ^ h[4] ^ h[6]

def main():
    n = int(input("Enter number: "))
    for b in range(1, n+1):
        d = list(map(int, input("Enter the
                data bits:").split())
        h = encode(d)
    print("Original data bits:", d)
    if(1<=pos<=7):
        h[pos] = 1
        print("Transmitted code with
                error induced")
    else print("Transmitted code:", h)
    d = [h[2], h[4], h[5], h[6]]
```

Output:

Enter number of 4bits data blocks

-- Black --

Enter 4 data bits --

original hamming code : 101

Encoded hamming code : 0110011

Transmitted code : 0100111

Recunmited code = 0100011.

-- bath --

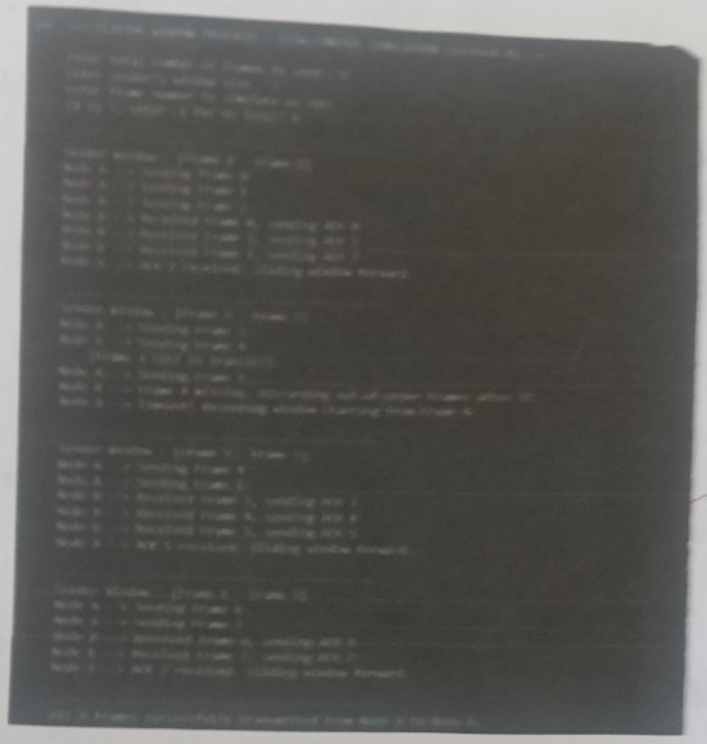
Enter 2data bits : 0110

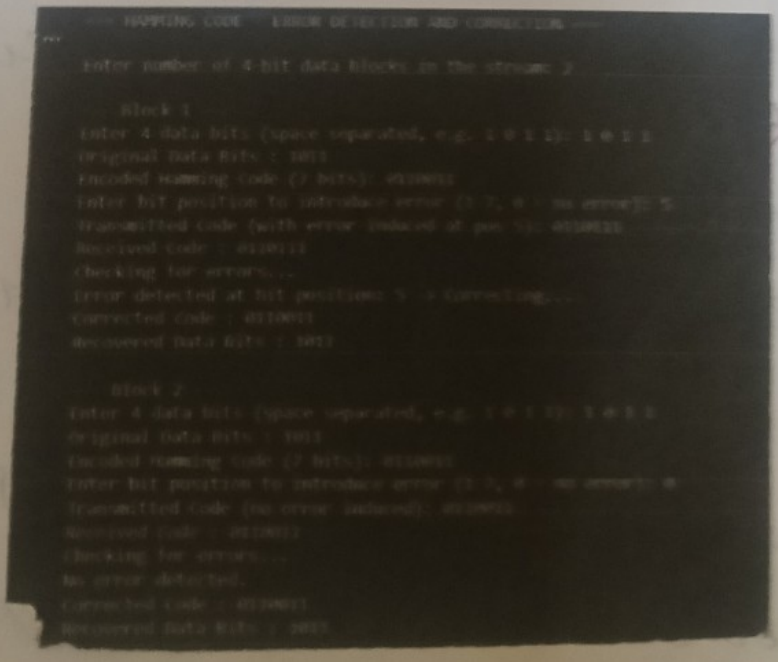
original data data = 0110

Encoded hamming code : 1100110

Transmitted code : 11000110

Recutiment code = 0110.





if __name__ == "__main__":
    main()

Result:

The program to implement error detection and correction using the Hamming code accept was successfully and verified